Contingent Claims Without Asset Representation

Abstract

Contingent claims are naturally described as state-contingent exposures over possible outcomes. In practice, however, financial systems usually represent such positions indirectly through portfolios of tradable assets. This induces a many-to-one mapping from portfolios to economic exposure, so that distinct portfolios and distinct asset-level trades can correspond to the same underlying economic state transition. We define a system in which positions are represented directly as exposure vectors in $\mathbb{R}^n$ and transactions operate directly in exposure space. Under this representation, economic state has a unique canonical form and economically equivalent trades collapse to the same update rule. The result is a transaction model in which netting is internal to the representation itself, rather than enforced by reconciling equivalent asset-level descriptions after the fact.

Keywords

prediction markets, DeFi, contingent claims, mechanism design

1. Introduction

Financial systems trade instruments whose value depends on uncertain future outcomes. These include prediction market shares, options, insurance contracts, and other contingent claims [1, 2, 3]. In economic theory, such positions are naturally described as *state-contingent exposures*: vectors specifying the amount of value held in each possible outcome [4, 5, 6].

In practice, however, financial systems do not represent positions directly in this form. Instead, they represent positions as portfolios of tradable assets, each with its own payoff profile, and the economic exposure of a participant is then derived from those holdings. This indirect architecture is standard across modern asset-pricing and market-microstructure frameworks [6, 7, 2]. It enables decentralised trade, but introduces a structural property: the mapping from asset portfolios to exposure is generally many-to-one.

As a consequence, asset-based representations exhibit *representational duplication*. Distinct portfolios can correspond to the same underlying exposure, and distinct trades can induce the same change in that exposure. These duplications arise from linear dependencies between asset payoffs and are independent of any particular market design.

Because economically equivalent states and trades can be represented in multiple ways, systems built on asset representations must maintain consistency between them. In practice, this is achieved through a combination of mechanisms such as matching, conversion, clearing, and arbitrage, which act to enforce equivalence between different asset-level descriptions of the same exposure [7, 8, 9, 10].

This paper isolates this representational property and studies an alternative approach. We define a system in which positions are represented directly as exposure vectors in $\mathbb{R}^n$, and trades are implemented as direct updates in exposure space. We show that, under this representation, both state and transaction representations become canonical: each economic position corresponds to a unique stored state, and each trade corresponds to a unique exposure update.

The contribution of the paper is therefore not to redefine contingent claims, but to define a representation and transaction model that operates directly on exposure as the primitive system state. This removes the

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duplication induced by asset-based representations at the level of state and updates, and eliminates the need to reconcile multiple asset-level descriptions of the same economic object within the representation itself.

This issue is particularly relevant in decentralised prediction-market systems, where contingent claims are implemented as tradable on-chain instruments and exchanged through explicit market mechanisms [1, 11, 12].

2. Exposure and Asset Representation

Let $\Omega = \{1, \dots, n\}$ denote the set of possible outcomes.

A participant's economic position is determined by how much value they hold in each outcome. We represent this as an exposure vector

$$\boldsymbol{e} \in \mathbb{R}^n,$$

where e_k denotes the value of the position in outcome $k \in \Omega$.

This is the standard state-contingent representation of a contingent claim in the Arrow–Debreu sense [4**?** , 6]. Two positions with the same exposure vector are economically indistinguishable, since they imply the same realised value in every outcome.

In an asset-based system, however, positions are not stored directly as exposure vectors. Instead, they are represented as portfolios of tradable assets.

Let there be m assets, each with exposure vector

$$\boldsymbol{x}^{(j)} \in \mathbb{R}^n, \qquad j = 1, \dots, m.$$

Collect these into the matrix

$$X \in \mathbb{R}^{n \times m},$$

whose jth column is $\boldsymbol{x}^{(j)}$.

A participant then holds a portfolio

$$\boldsymbol{q} \in \mathbb{R}^m,$$

which induces exposure

$$\boldsymbol{e} = X\boldsymbol{q}.$$

Thus, the economic object of interest is the exposure vector $\boldsymbol{e}$, while the system stores a portfolio $\boldsymbol{q}$ together with a fixed payoff map X.

Interpretation. Asset-based systems do not operate directly on exposure. Instead, they operate on portfolios, and exposure is derived as a linear image of those portfolios.

In the next section, we show that this representation is generally non-unique: multiple portfolios can induce the same exposure, and multiple portfolio updates can induce the same change in exposure.

3. Representational Duplication

In an asset-based system, exposure is given by

$$\boldsymbol{e} = X\boldsymbol{q},$$

where $\boldsymbol{q} \in \mathbb{R}^m$ is a portfolio and $X \in \mathbb{R}^{n \times m}$ is the asset payoff matrix.

The economic state is therefore $\boldsymbol{e}$, while the system stores $\boldsymbol{q}$. This representation is generally non-unique.

3.1. State Duplication

State duplication occurs when distinct portfolios induce the same exposure.

Formally, the representation is non-unique if there exist

$$\boldsymbol{q}_1 \neq \boldsymbol{q}_2 \quad \text{such that} \quad X\boldsymbol{q}_1 = X\boldsymbol{q}_2.$$

Structural source. This non-uniqueness arises whenever the columns of X are linearly dependent, so that some assets can be replicated as linear combinations of others.

In contingent-claim systems, this dependence arises naturally from the relationship between an underlying payoff and its contingent decomposition.

Consider a state-independent asset (cash)

$$\$ = (1, \ldots, 1),$$

together with outcome-contingent claims

$$\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n,$$

where $\boldsymbol{e}_k$ pays one unit in outcome k and zero otherwise.

These satisfy

$$\$ = \sum_{k=1}^{n} \boldsymbol{e}_k,$$

so the underlying payoff can be replicated exactly by combining contingent claims.

Thus, once both the underlying asset and its contingent claims are tradable as assets, the payoff matrix X is necessarily linearly dependent.

Interpretation. Representational duplication arises as soon as both an underlying payoff and tradable contingent claims on that payoff are present in the system. Multiple distinct portfolios then correspond to the same exposure vector $\boldsymbol{e}$.

Example (binary market). Let $\Omega = \{Y, N\}$ and define assets

$$\boldsymbol{x}^{(Y)} = (1, 0), \quad \boldsymbol{x}^{(N)} = (0, 1), \quad \boldsymbol{x}^{(\$)} = (1, 1).$$

Then

$$\boldsymbol{x}^{(\$)} = \boldsymbol{x}^{(Y)} + \boldsymbol{x}^{(N)},$$

so holding one unit of cash is equivalent to holding one unit of each outcome claim.

Example (three outcomes with complementary claims). Let $\Omega = \{R, G, B\}$ and define assets

$$\boldsymbol{x}^{(Y_R)} = (1, 0, 0), \quad \boldsymbol{x}^{(Y_G)} = (0, 1, 0), \quad \boldsymbol{x}^{(Y_B)} = (0, 0, 1),$$

and complementary claims by

$$\boldsymbol{x}^{(N_R)} = \boldsymbol{x}^{(\$)} - \boldsymbol{x}^{(Y_R)} = (0, 1, 1).$$

Then

$$\boldsymbol{x}^{(N_R)} = \boldsymbol{x}^{(Y_G)} + \boldsymbol{x}^{(Y_B)},$$

so holding one unit of NO-R is equivalent to holding one unit of YES-G and YES-B.

Thus, distinct portfolios again represent the same exposure.

3.2. Transaction Duplication

Transaction duplication occurs when distinct portfolio updates induce the same change in exposure.

A trade is represented as

$$\Delta \boldsymbol{q} \in \mathbb{R}^m, \quad \Delta \boldsymbol{e} = X \Delta \boldsymbol{q}.$$

The representation of trades is non-unique if there exist

$$\Delta \boldsymbol{q}_1 \neq \Delta \boldsymbol{q}_2 \quad \text{such that} \quad X \Delta \boldsymbol{q}_1 = X \Delta \boldsymbol{q}_2.$$

Thus, multiple asset-level trades correspond to the same economic trade $\Delta \boldsymbol{e}$.

Example (binary claims). Let

$$\boldsymbol{x}^{(Y_k)} = \boldsymbol{e}_k, \quad \boldsymbol{x}^{(N_k)} = \$ - \boldsymbol{e}_k.$$

Buying one unit of Y_k at price p induces

$$\Delta \boldsymbol{e} = \boldsymbol{e}_k - p\$.$$

Selling one unit of N_k at price $1 - p$ induces

$$\Delta \boldsymbol{e} = (1-p)\$ - (\$ - \boldsymbol{e}_k) = \boldsymbol{e}_k - p\$.$$

These are distinct asset-level trades, but they produce the same exposure update.

Interpretation. Just as multiple portfolios can represent the same state, multiple trades can represent the same change in state. This duplication arises from the same linear dependence structure in X. Thus, buying YES for price p and selling NO for price 1-p are distinct asset-level trades that produce the same exposure update.

3.3. Summary

Asset-based representations introduce two forms of non-uniqueness:

- **state duplication**, where multiple portfolios represent the same exposure;
- **transaction duplication**, where multiple trades produce the same exposure update.

These are not artifacts of a particular market design. They arise structurally from representing contingent claims as portfolios of assets defined over a common underlying payoff. Whenever both an underlying asset and its contingent claims are tradable, the representation is necessarily redundant.

4. Exposure-Based Representation

4.1. State Representation

In the exposure-based system, the protocol records a participant's economic position directly as exposure.

For each participant i and market m:

$$\boldsymbol{e}_{i,m} \in \mathbb{R}^n,$$

where $(e_{i,m})_k$ is the value held in outcome $k \in \Omega_m$.

The system enforces a feasibility constraint:

$$(e_{i,m})_k \geq 0 \quad \forall k, m.$$

That is, exposure must be nonnegative in every outcome of every market.

4.2. Canonical Decomposition

Define the cash vector:

$$\$ = (1, 1, \ldots, 1).$$

Each exposure vector admits a decomposition:

$$\boldsymbol{e}_{i,m} = c_i \$ + \boldsymbol{r}_{i,m},$$

where

$$c_i = \min_{m,k}(e_{i,m})_k, \qquad \boldsymbol{r}_{i,m} = \boldsymbol{e}_{i,m} - c_i \$.$$

By construction:

$$(\boldsymbol{r}_{i,m})_k \geq 0 \quad \forall k, m, \qquad \min_{m,k}(\boldsymbol{r}_{i,m})_k = 0.$$

The scalar c_i is the minimum value of the participant's exposure across all outcomes and all markets.

4.3. Constraint Reduction

The feasibility condition requires:

$$(e_{i,m})_k \geq 0 \quad \forall k, m.$$

Under the decomposition:

$$(e_{i,m})_k = c_i + (r_{i,m})_k, \quad \text{with} \quad (r_{i,m})_k \geq 0.$$

Therefore:

$$\min_{m,k}(e_{i,m})_k = c_i.$$

It follows that:

$$\boldsymbol{e}_{i,m} \geq 0 \; \forall m \quad \Longleftrightarrow \quad c_i \geq 0.$$

Interpretation. A collection of vector inequality constraints is reduced to a single scalar invariant.

The system does not need to check feasibility across all outcomes and all markets. It is sufficient to track and enforce:

$$c_i \geq 0.$$

4.4. State Transition Rule

A trade updates exposure directly.

For participant i:

$$\boldsymbol{e}_{i,m} \to \boldsymbol{e}_{i,m} + \Delta \boldsymbol{e}_m.$$

The protocol applies the following procedure:

1. Apply the exposure update across affected markets.
2. Compute the new minimum component:
$$c_i' = \min_{m,k} (e_{i,m}')_k.$$
3. If $c_i' < 0$, revert the transaction.
4. Otherwise, rewrite the state in canonical form.

4.5. Canonicalisation

After an update, define:

$$\mu = \min_{m,k} (r_{i,m}')_k.$$

Then update:

$$\boldsymbol{r}_{i,m}' \leftarrow \boldsymbol{r}_{i,m}' - \mu \$, \qquad c_i \leftarrow c_i + \mu.$$

This preserves $\boldsymbol{e}_{i,m}$ while restoring the canonical condition:

$$\min_{m,k} (\boldsymbol{r}_{i,m})_k = 0.$$

Canonicalisation changes only representation, not economic state.

4.6. Summary

In the exposure-based system, feasibility is determined by a single scalar invariant.

A participant's position must satisfy

$$\min_{k \in \Omega} (e_i)_k \geq 0,$$

which is equivalent to

$$c_i \geq 0.$$

Thus, a high-dimensional solvency constraint over all outcomes reduces exactly to a single scalar condition. Trades are accepted if and only if this condition is preserved.

5. Exposure Transactions

In the exposure-based system, a trade is defined directly as a transfer of exposure between participants.

For participants i and j, a trade is given by a vector

$$\Delta \boldsymbol{e} \in \mathbb{R}^n$$

such that

$$\boldsymbol{e}_i \to \boldsymbol{e}_i + \Delta \boldsymbol{e}, \quad \boldsymbol{e}_j \to \boldsymbol{e}_j - \Delta \boldsymbol{e}.$$

The same vector $\Delta \boldsymbol{e}$ defines the trade for both participants.

Conservation. This update preserves total exposure:

$$(\boldsymbol{e}_i + \Delta\boldsymbol{e}) + (\boldsymbol{e}_j - \Delta\boldsymbol{e}) = \boldsymbol{e}_i + \boldsymbol{e}_j.$$

A trade therefore reallocates exposure between participants without changing aggregate exposure in the system.

Interpretation. Trades are represented directly in exposure space. Each economic trade corresponds to a unique update vector $\Delta\boldsymbol{e}$, independent of any asset-level description.

Example. Taking one unit of exposure in outcome k at price p is represented by

$$\Delta\boldsymbol{e} = \boldsymbol{e}_k - p\$.$$

This vector fully specifies the economic effect of the trade. Asset-level distinctions such as buying YES or selling NO do not arise, as they correspond to the same $\Delta\boldsymbol{e}$ and are therefore not separately represented.

6. Conclusion

Standard financial implementations represent contingent claims indirectly through portfolios of tradable assets. Because the mapping from portfolios to state-contingent exposure is generally many-to-one, the same economic position can appear under multiple portfolio descriptions, and the same economic trade can appear under multiple asset-level transactions. This creates representational duplication at the level of both stored state and state transition.

We have defined an alternative model in which the primitive system object is exposure itself. Positions are stored directly as exposure vectors, rewritten in canonical form, and updated by direct exposure transfers. In this representation, economically equivalent states coincide by construction, and economically equivalent trades are represented by the same update vector.

The main contribution is therefore representational. Rather than encoding exposure indirectly and relying on additional mechanisms to reconcile equivalent portfolio descriptions, the system stores the economic object directly. This removes duplication within the representation itself and provides a canonical state-transition model for contingent claims.

A. Appendix: State Transitions Between Two Participants

Setup. We consider two participants, Alice and Bob, trading in a market with outcomes

$$\Omega = \{R, G, B\}.$$

Each participant's state is represented by:

$$(c, \boldsymbol{r}, \boldsymbol{e}), \quad \text{with} \quad \boldsymbol{e} = c\$ + \boldsymbol{r}, \quad \boldsymbol{r} \geq 0, \quad \min(\boldsymbol{r}) = 0,$$

where

$$\$ = (1, 1, 1).$$

Convention. Each transition is defined by a single exposure update $\Delta \boldsymbol{e}$ applied to Alice:

$$\boldsymbol{e}_A \to \boldsymbol{e}_A + \Delta \boldsymbol{e}, \qquad \boldsymbol{e}_B \to \boldsymbol{e}_B - \Delta \boldsymbol{e}.$$

Scalar invariant. Solvency reduces to a single scalar condition:

$$\boldsymbol{e} \geq 0 \quad \Longleftrightarrow \quad c \geq 0.$$

Initial state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	10	$(0,0,0)$	$(10,10,10)$
Bob	5	$(0,0,0)$	$(5,5,5)$

A.1. Transition 1: Cash transfer

Alice sends Bob 2

$$\Delta \boldsymbol{e} = (-2,-2,-2)$$

Initial state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	10	$(0,0,0)$	$(10,10,10)$
Bob	5	$(0,0,0)$	$(5,5,5)$

Apply update.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	10	$(0,0,0)$	$(8,8,8)$
Bob	5	$(0,0,0)$	$(7,7,7)$

Canonical form (unchanged).

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	8	$(0,0,0)$	$(8,8,8)$
Bob	7	$(0,0,0)$	$(7,7,7)$

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A.2. Transition 2: Directional transfer (3 YES-G)

Alice sends Bob 3 Yes Green

Initial state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	8	$(0,0,0)$	$(8,8,8)$
Bob	7	$(0,0,0)$	$(7,7,7)$

$$\Delta \boldsymbol{e} = (0,-3,0)$$

Apply update.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	8	$(0,-3,0)$	$(8,5,8)$
Bob	7	$(0,3,0)$	(7,10,7)

Canonicalisation.

$$\mu = \min(\boldsymbol{r}_A) = -3$$

$$\boldsymbol{r}_A \to \boldsymbol{r}_A - \mu \mathbf{\$}, \quad c_A \to c_A + \mu$$

Final state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	5	$(3,0,3)$	$(8,5,8)$
Bob	7	$(0,3,0)$	$(7,10,7)$

A.3. Transition 3: Transfer of 1 NO-R

Alice sends Bob 1 No R

Initial state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	5	$(3,0,3)$	$(8,5,8)$
Bob	7	$(0,3,0)$	$(7,10,7)$

$$\Delta \boldsymbol{e} = (0,-1,-1)$$

Apply update.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	5	$(3,-1,2)$	$(8,4,7)$
Bob	7	$(0,4,1)$	$(7,11,8)$

Canonicalisation. Alice

$$\mu = -1$$

$$\boldsymbol{r}_A \to (4, 0, 3), \quad c_A \to 4$$

Final state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	4	$(4, 0, 3)$	$(8, 4, 7)$
Bob	7	$(0, 4, 1)$	$(7, 11, 8)$

A.4. Transition 4: Priced trade (YES-R at 0.2)

Alice sells 1 YES-R to Bob for 0.2

Initial state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	4	$(4, 0, 3)$	$(8, 4, 7)$
Bob	7	$(0, 4, 1)$	$(7, 11, 8)$

$$\Delta c = +0.2, \quad \Delta \boldsymbol{r} = (-1, 0, 0)$$

$$\Delta \boldsymbol{e} = (-0.8, 0.2, 0.2)$$

Final state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	4.2	$(3, 0, 3)$	$(7.2, 4.2, 7.2)$
Bob	7.8	$(0, 3, 0)$	$(7.8, 10.8, 7.8)$

A.5. Attempted Transition: Invalid trade

Alice tries to buy 10 YES-B from Bob for \$6.

$$\Delta c = -6, \quad \Delta \boldsymbol{r} = (0, 0, 10)$$

$$\Delta \boldsymbol{e} = (-6, -6, 4)$$

Initial state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	4	$(0, 2, 0)$	$(4, 6, 4)$
Bob	9	$(2, 0, 2)$	$(11, 9, 11)$

Apply update.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	-2	$(0, 2, 10)$	$(-2, 0, 8)$
Bob	15	$(2, 0, -8)$	$(17, 15, 7)$

Alice already has $c_A = -2 < 0$. Bob's state is not yet canonical since $\boldsymbol{r}_B$ has a negative component.

Canonicalisation. For Bob:

$$\mu_B = \min(\boldsymbol{r}_B) = -8$$

$$\boldsymbol{r}_B \to \boldsymbol{r}_B - \mu_B \mathbf{\$} = (2, 0, -8) + (8, 8, 8) = (10, 8, 0)$$

$$c_B \to c_B + \mu_B = 15 - 8 = 7$$

Alice is already in canonical form since

$$\min(\boldsymbol{r}_A) = 0.$$

Canonical state.

User	c	$\boldsymbol{r}$	$\boldsymbol{e}$
Alice	-2	$(0, 2, 10)$	$(-2, 0, 8)$
Bob	7	$(10, 8, 0)$	$(17, 15, 7)$

Failure condition. For Alice:

$$c_A = -2 < 0$$

Hence

$$\min(\boldsymbol{e}_A) = -2 < 0,$$

so Alice's exposure is negative in at least one outcome. The transaction therefore violates the feasibility condition and reverts.

A.6. Transition Across Two Markets

We now consider a participant with exposure across two markets. The purpose of the example is to show that a trade in one market changes exposure in that market only, while solvency is still determined by a single global scalar c_A.

Let

$$\Omega_{m_1} = \{Y, N\}, \qquad \Omega_{m_2} = \{Y, N\},$$

and let Alice's state be represented by

$$(c_A, \boldsymbol{r}_{A,m_1}, \boldsymbol{r}_{A,m_2}),$$

with

$$\boldsymbol{e}_{A,m_1} = c_A \$ + \boldsymbol{r}_{A,m_1}, \qquad \boldsymbol{e}_{A,m_2} = c_A \$ + \boldsymbol{r}_{A,m_2}, \qquad \$ = (1,1).$$

Suppose initially

$$c_A = 5, \qquad \boldsymbol{r}_{A,m_1} = (2,0), \qquad \boldsymbol{r}_{A,m_2} = (0,3).$$

Then

$$\boldsymbol{e}_{A,m_1} = (7,5), \qquad \boldsymbol{e}_{A,m_2} = (5,8).$$

Market	c_A	$\boldsymbol{r}_{A,m}$	$\boldsymbol{e}_{A,m}$
m_1	5	$(2,0)$	$(7,5)$
m_2	5	$(0,3)$	$(5,8)$

Alice now executes a trade in market m_1 only, with exposure update

$$\Delta \boldsymbol{e}_{A,m_1} = (-3,1).$$

So the new exposure in market m_1 is

$$\boldsymbol{e}'_{A,m_1} = (7,5) + (-3,1) = (4,6).$$

At this point market m_2 has not been traded, so its residual vector remains

$$\boldsymbol{r}_{A,m_2} = (0,3).$$

To determine whether Alice remains solvent, we now recompute the global minimum component across all markets:

$$c'_A = \min\{4,6,5,8\} = 4.$$

Since

$$c'_A = 4 \geq 0,$$

Alice remains solvent.

We now rewrite the state in canonical form using the updated global scalar $c'_A = 4$.

For market m_1,

$$\boldsymbol{r}'_{A,m_1} = \boldsymbol{e}'_{A,m_1} - c'_A \$ = (4,6) - (4,4) = (0,2).$$

For market m_2, the residual vector is unchanged:

$$\boldsymbol{r}'_{A,m_2} = \boldsymbol{r}_{A,m_2} = (0,3).$$

But because c_A is global, the exposure vector in market m_2 is now written relative to the new scalar:

$$\boldsymbol{e}'_{A,m_2} = c'_A\$ + \boldsymbol{r}'_{A,m_2} = (4,4) + (0,3) = (4,7).$$

Thus the final canonical state is:

Market	c_A	$\boldsymbol{r}_{A,m}$	$\boldsymbol{e}_{A,m}$
m_1	4	$(0,2)$	$(4,6)$
m_2	4	$(0,3)$	$(4,7)$

The important point is that the trade occurs only in market m_1, so only $\boldsymbol{e}_{A,m_1}$ is updated directly by the transaction. Solvency is then checked by recomputing the single global scalar c_A. Market m_2 is not traded, so its residual vector $\boldsymbol{r}_{A,m_2}$ is unchanged. However, because c_A is global, the exposure vector $\boldsymbol{e}_{A,m_2}$ changes when the state is rewritten in canonical form.

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